

SWARAJ SHINDE

Data Scientist

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Professional Summary

Data scientist professional with experience in generative AI, multi-agent systems, machine learning, data science and data analytics. Skilled in Python for ETL, model development, and exploratory analysis. Currently building LLM-based applications and seeking a role focused on production-ready AI solutions.

Skills

- **Generative AI & LLMs:** RAG (Retrieval-Augmented Generation), Tool / Function Calling, Prompt Engineering, AI Agents (ReAct, Plan-and-Execute), Fine-tuning LLMs, Embeddings & Vector Search, Structured Outputs, LLM Evaluation (RAGAS), LoRA, PEFT
- **Frameworks & Orchestration:** LangChain, LangGraph, Langfuse, Langsmith, HuggingFace Transformers, PyTorch, Scikit-learn, FastAPI, Streamlit
- **ML & Data Science:** Machine Learning, NLP, Time Series Analysis, Recommendation Systems, Classification, Regression, Clustering, EDA, Feature Engineering, ETL Pipelines, Model Evaluation Metrics, Predictive Analytics
- **Programming & Databases:** Python, Pandas, SQL, NumPy, NLTK, ChromaDB, FAISS, Pinecone
- **MLOps & Deployment:** Docker, MLflow, REST APIs, Git, GitHub
- **Tools & Platforms:** Jupyter Notebook, VS Code, Claude, Power BI, Microsoft Excel

Professional Experience

Shivrai Technology Pvt. Ltd, Pune | Data Scientist

Dec 24 – July 2026

- **Project: Agentic AI-Driven Expense Estimation System**
 - Developed Agentic AI expense-estimation agents using OpenAI API(GPT-4o) with rule-based and 5-tool-based cost modelling to automate cost analysis for farming operations, reducing manual effort by 90%.
 - Supported data-driven financial planning for 10,000+ farmers through scalable cost modelling across labour, fertilizers, and seasonal variables.
 - **Key Achievement:**
Delivered a production-ready Agentic AI system that automated complex cost estimation workflows for clients, ensuring scalable, reliable, and data-driven financial planning.
- **Project: RAG-Based Knowledge QA Chatbot**
 - Built and deployed a Hybrid RAG chatbot with vector search and embedding-based retrieval, delivering context-aware answers and handling 10,000+ queries per month.
 - Automated support query resolution, reducing support workload by 40% and cutting response time.
 - **Key Achievement:**
Built a scalable RAG pipeline that consistently retrieved high-quality context, enabling reliable AI-driven knowledge assistance.
- **Project: ML-Based Sales Forecasting & Inventory Optimization**
 - Developed time-series sales forecasting models using Random Forest, LightGBM, and Decision Tree etc. for weekly and monthly vendor orders, identifying trends and seasonal patterns to improve demand forecasting accuracy by 15%.
 - Supported supply chain and procurement decisions, improving inventory planning and reducing wastage by 70%.
 - **Key Achievement:**
Delivered a scalable forecasting solution that aligned inventory planning with demand trends, directly improving operational efficiency and margins.

- **Project: LiDAR-Based Roadside Infrastructure Detection & Computer Vision System**
 - Performed 2,000+ image annotations using LabelMe and CVAT, significantly improving dataset quality and enhancing object detection model accuracy.
 - Trained YOLOv8-based Computer Vision models to detect road-side and high-infrastructure objects such as poles, street lights, lamps, dividers, and related infrastructure components using LiDAR and image-based datasets.
 - **Key Achievement:**
Developed an efficient infrastructure detection pipeline capable of improving automated road-scene understanding and supporting smart transportation and autonomous driving applications.

Personal Projects

- SmartTrip AI: | [Link](#) |
- Feb 26
- Built an agentic AI trip planning system using LLMs with 4 integrated tools/APIs for itinerary generation, hotel recommendations, route optimization, and travel assistance using the Ollama model gpt-oss:120b, implementing multi-tool orchestration, prompt engineering, and real-time API integration for personalized travel planning.
- PharmGPT Chatbot: | [Link](#) |
- March 26
- Deployed a system using Gemini 2.5 Flash-Lite for text generation and Stability AI's Flux model for generating images of various yoga activities. The models were primarily trained on healthcare and pharmaceutical domain data.

Education

- Dr. Babasaheb Ambedkar Technological University, Maharashtra
- Jun 20 - Jun 24
- B-Tech in Computer Science and Engineering | CGPA: 7.5

Publications

- Placement Cell System: AI powered application for placement department of college. | [Link](#) |
- | Published by IJIRSET International Journal (Paper ID – 15680/IJIRSET)

Certifications

- | Build Multimodal Generative AI Applications | [Certificate](#) |
 - | Data Science Specialization. | [Certificate](#) |
- Jun 26
- Mar 24